Course: 65CS037 – Machine Learning

Task - Classification

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Course: 5CS037/HJ1: Concepts and Technologies of AI

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Classification of High Obesity Prevalence Using Behavioral Risk Factor Surveillance System Data

Abstract

Purpose: The goal of this report is to predict whether obesity prevalence is high (≥30%) or not using classification techniques.

Dataset: The dataset used is “Nutrition, Physical Activity, and Obesity – Behavioral Risk Factor Surveillance System (BRFSS)” from CDC. It contains 110,880 records and 33 columns. After cleaning, we used 19,078 rows for modeling. This dataset aligns with United Nations Sustainable Development Goal 3 (Good Health and Well-being) by helping understand and reduce obesity in adults.

Approach: We did Exploratory Data Analysis (EDA), built a Neural Network classifier, two classical models (Logistic Regression and Random Forest), did hyperparameter tuning, feature selection, and compared all models.

Key Results: Models were evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Neural Network gave the best result (Accuracy 0.8184, ROC-AUC 0.8970). Logistic Regression was the best classical model after tuning and feature selection (Accuracy 0.7403, F1-Score 0.8106).

Conclusion: Final models performed well to identify high-obesity groups. Neural Network was strongest, but classical models are simpler and useful for health planning.

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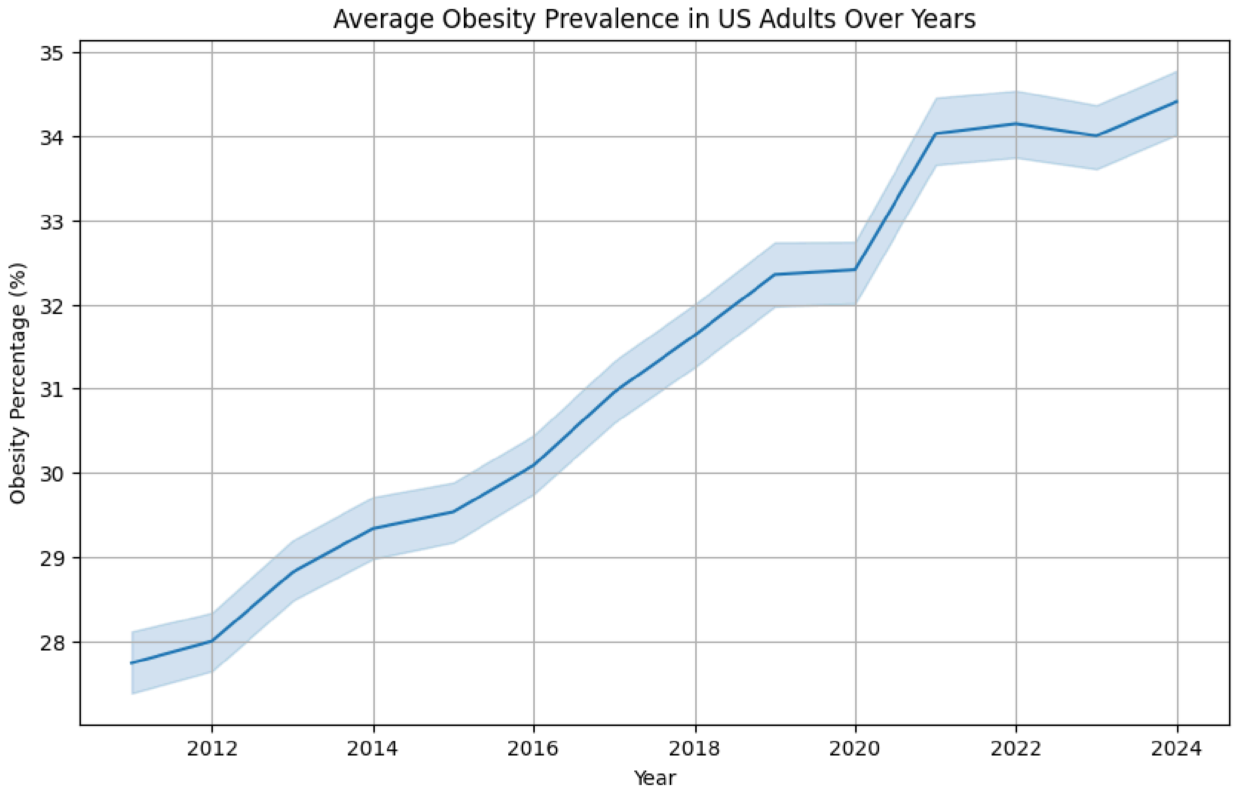


Figure 1: Obesity rate over years

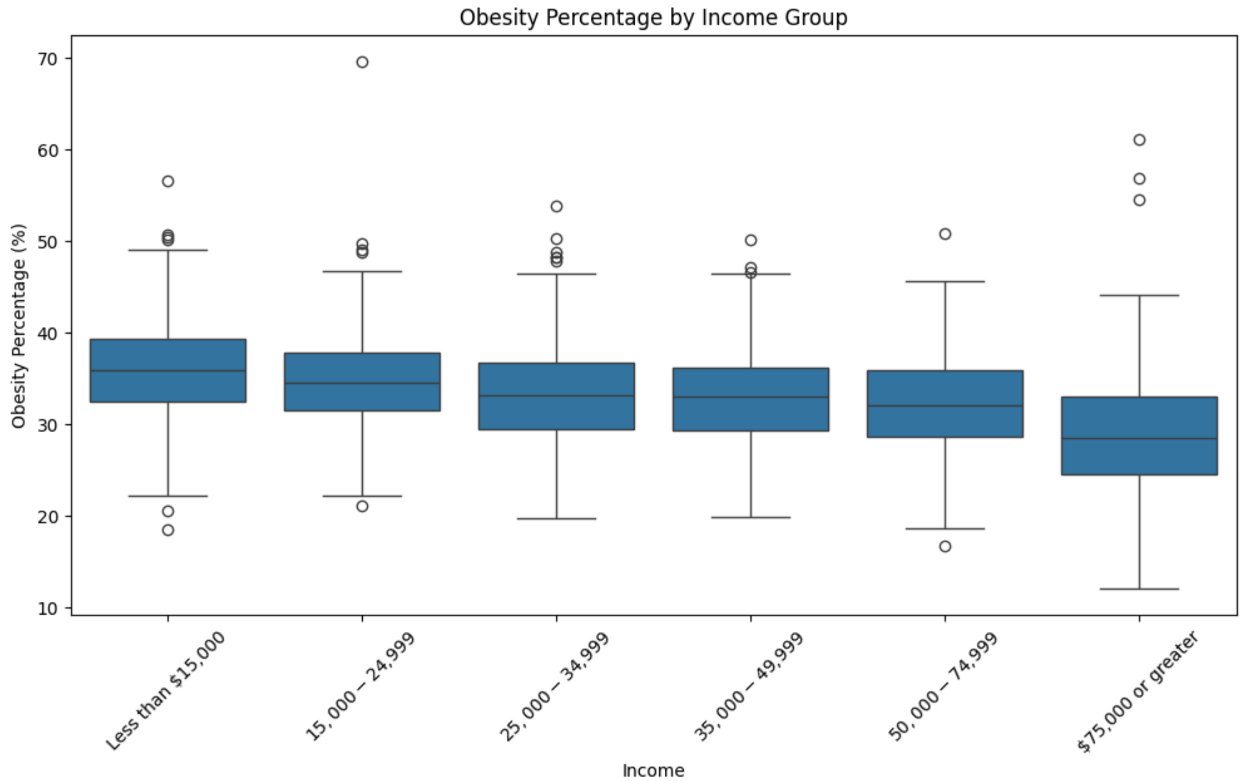


Figure 2: Obesity by income level

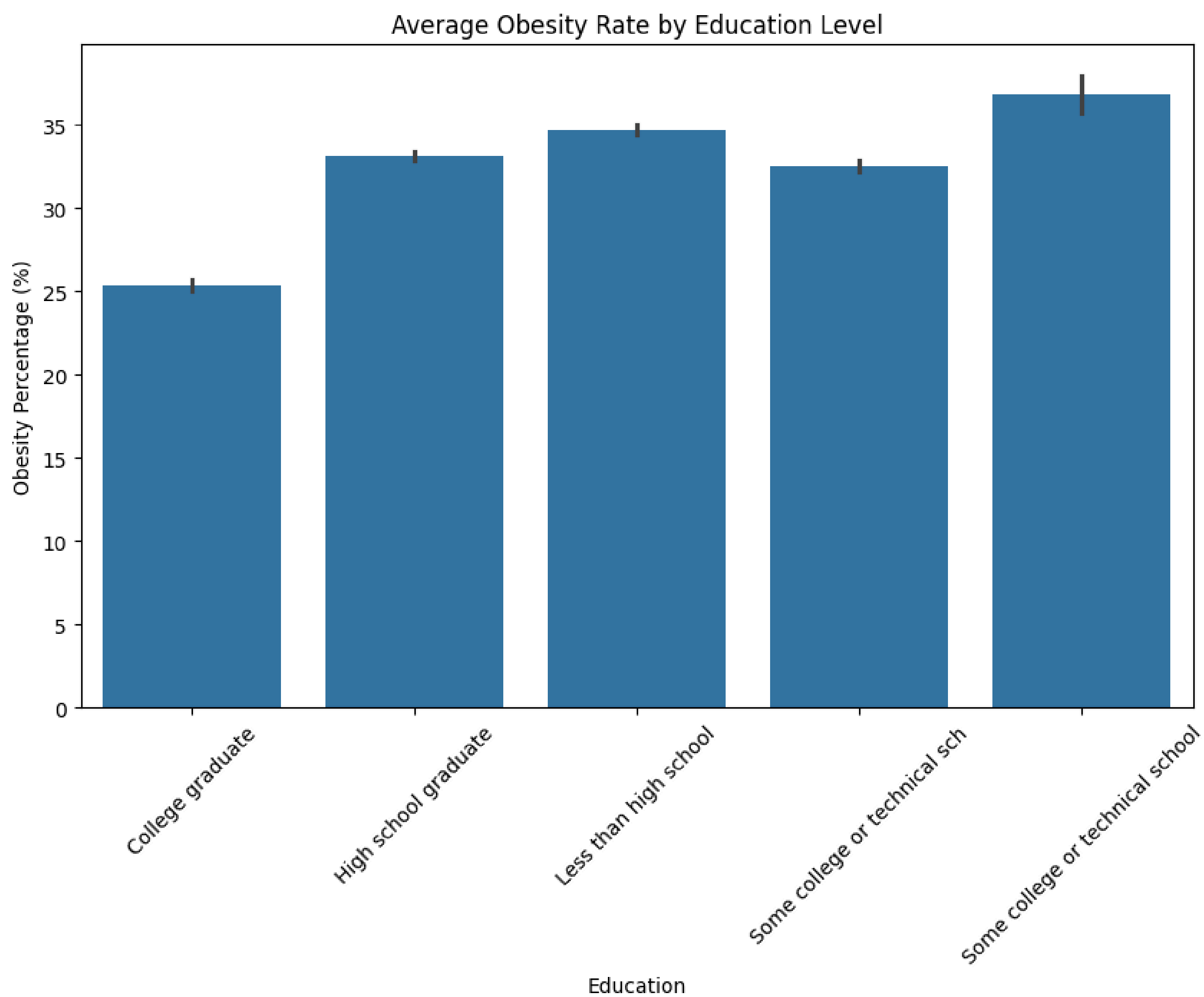


Figure 3: Obesity by education level

1. Introduction

1.1 Problem Statement

The goal of this project is to classify whether obesity prevalence in a group is high (≥30%) or not. This helps public health workers find high-risk groups and plan better programs.

1.2 Dataset

The dataset is “Nutrition, Physical Activity, and Obesity – Behavioral Risk Factor Surveillance System (BRFSS)” from CDC website (data.cdc.gov). It was created by Centers for Disease Control and Prevention (CDC) starting from 1984, with updates every year. We used data from 2011 to recent years.

This dataset supports SDG 3 (Good Health and Well-being) because obesity is a big risk for many diseases like diabetes and heart problems.

1.3 Objective

To build and compare classification models (Neural Network, Logistic Regression, Random Forest) to predict high obesity accurately using survey features.

2. Methodology

2.1 Data Preprocessing

We cleaned the data by removing rows without Data_Value. We focused on obesity question rows. We created a binary target: High_Obesity = 1 if Data_Value ≥ 30%, else 0. We scaled numeric features and one-hot encoded categorical features (Income, Education, Sex, etc.).

2.2 Exploratory Data Analysis (EDA)

We used histograms, bar plots, box plots, and line plots.

Key insights:

- Obesity rate is slowly increasing over years (around 28–32%).
- Lower income groups have higher obesity (often >30–35%).
- Less educated people have higher obesity rates.
- Southern states like Mississippi have highest obesity.
- Small sample sizes give wide confidence intervals (less reliable).

2.3 Model Building

Neural Network (MLP):

- Architecture: Input → 128 (ReLU + Dropout 0.3) → 64 (ReLU + Dropout 0.3) → 32 (ReLU + Dropout 0.2) → 1 (Sigmoid)
- Loss: Binary Cross-Entropy
- Optimizer: Adam (lr=0.001)
- Trained for 50 epochs

Two Classical ML Models:

1. **Logistic Regression** – simple and interpretable
2. **Random Forest Classifier** – strong ensemble method

We split data 80% train (15,262 samples) and 20% test (3,816 samples) with stratification.

2.4 Model Evaluation

We used these metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

2.5 Hyper-parameter Optimization

For classical models only.

Logistic Regression:

- Used GridSearchCV (5-fold)
- Tuned C, class_weight, solver
- **Best parameters**: C=100.0, class_weight='balanced', solver='liblinear'

- **Best CV ROC-AUC**: 0.8864

Random Forest Classifier:

- Used RandomizedSearchCV (20 iterations, 5-fold)
- Tuned n_estimators, max_depth, min_samples_split, etc.
- **Best parameters**: n_estimators=200, min_samples_split=10, min_samples_leaf=1, max_features='log2', max_depth=None, class_weight='balanced'
- **Best CV ROC-AUC**: 0.8964

2.6 Feature Selection

We used embedded method – Random Forest feature importance.

Selected top features (importance > mean): 11 features.

Selected features:

- Sample_Size
- Income_Data not reported
- Education_Some college or technical sch
- Race/Ethnicity_Asian
- Age(years)_35 - 44
- Age(years)_45 - 54
- Age(years)_55 - 64
- Age(years)_nan
- LocationAbbr_CO
- LocationAbbr_HI
- LocationAbbr_MA

Justification: These features explain most of the prediction. Sample_Size is most important (reliability of data). Income, education, race, age, and some states are known risk factors for obesity. Using only these makes models simpler and faster without losing much accuracy.

3. Results and Conclusion

Table 4: Comparison of Final Models

Model	Features	CV Score	Accuracy	Precision	Recall	F1-Score
Logistic Regression (Final)	Selected (11)	0.8864 (ROC-AUC)	0.7403	0.7270	0.9158	0.8106
Random Forest Classifier (Final)	Selected (11)	0.8964 (ROC-AUC)	0.6740	0.7364	0.7205	0.7284

3.1 Key Findings

Neural Network gave best result (Accuracy 0.8184, ROC-AUC 0.8970).
Logistic Regression was best among classical models after tuning and feature selection.

Using only 11 features still gave good performance.

3.2 Final Model

Logistic Regression (Final) with selected 11 features is the best classical model (Accuracy 0.7403, F1-Score 0.8106).

3.3 Challenges

- Many missing values in stratification columns
- Overflow error in neural network log-target
- Long tuning time for Random Forest

3.4 Future Work

- Try XGBoost or LightGBM
- Add more features (e.g., diet details)
- Use deep learning with better layers
- Test on new data from other countries

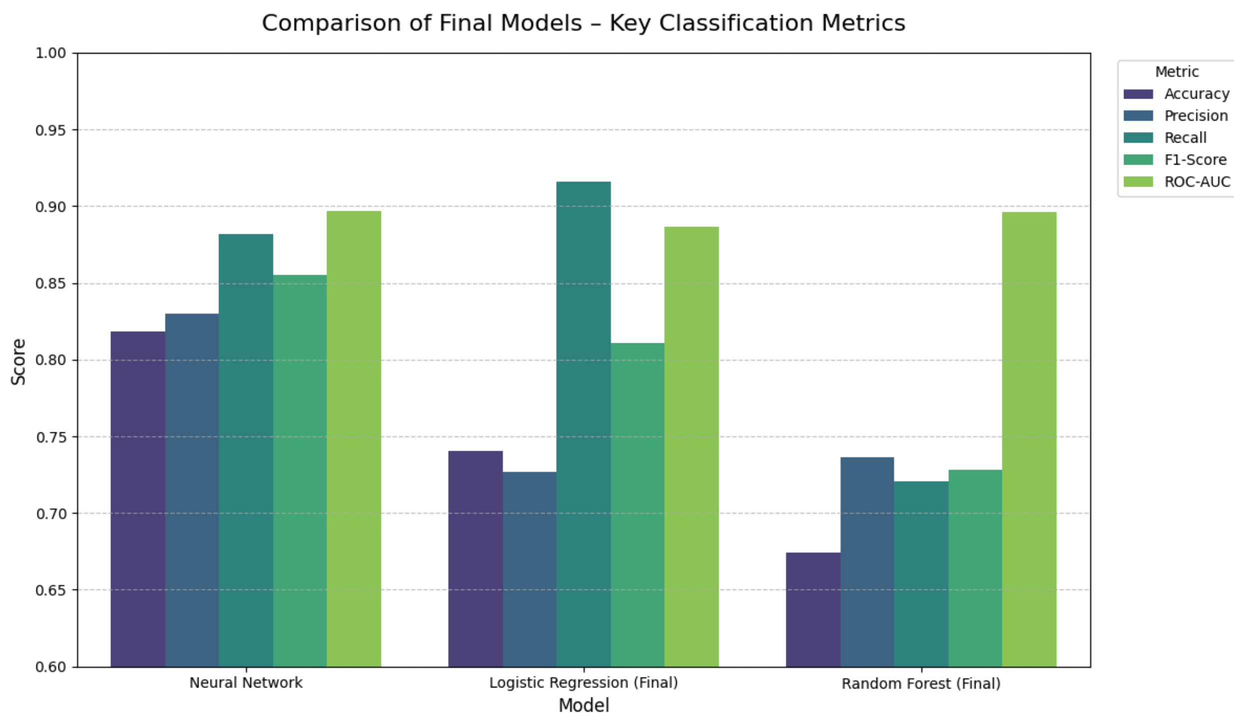
4. Discussion

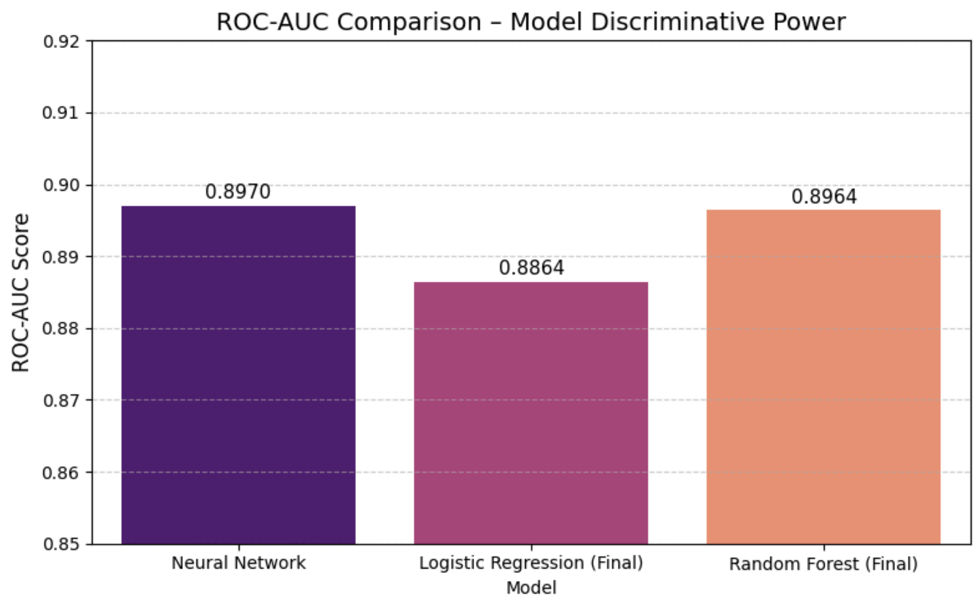
4.1 Model Performance

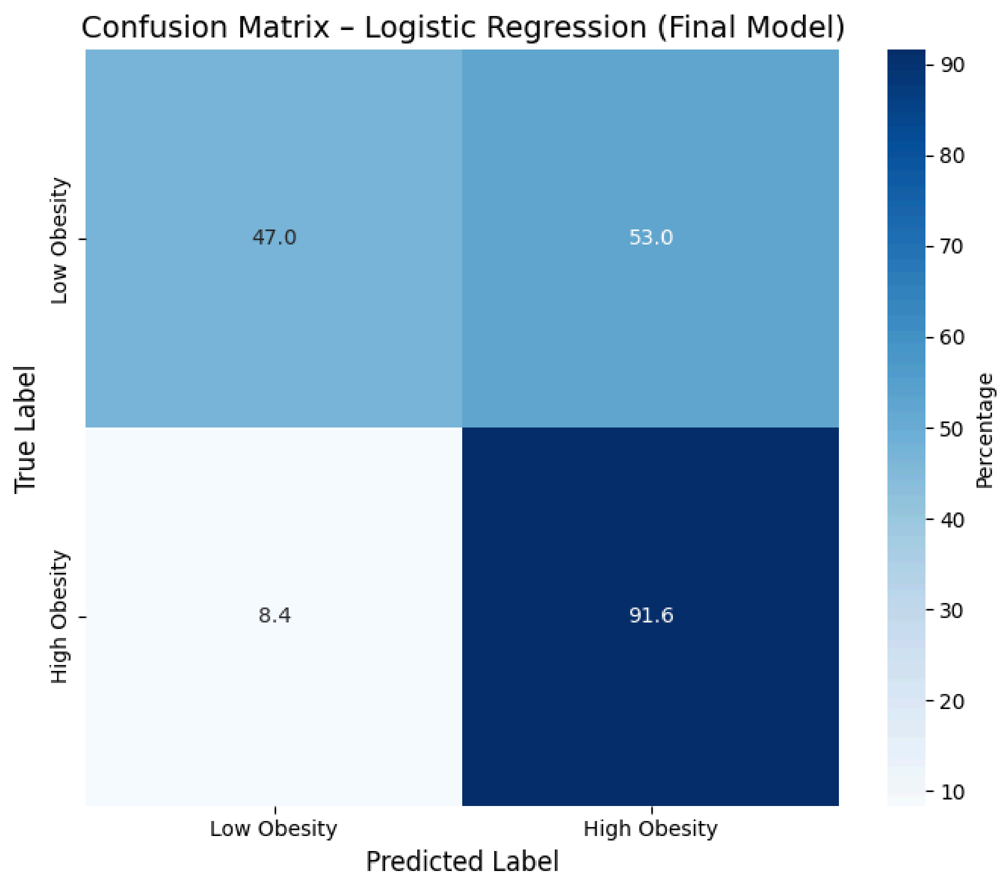
Neural Network had highest accuracy and ROC-AUC.

Logistic Regression performed better than Random Forest after feature selection.

Models are good at finding high-obesity groups (high recall for positive class).




ROC-AUC Comparison – Model Discriminative Power
ROC-AUC Score
0.92
0.91
0.90
0.89
0.88
0.87
0.86
0.85
0.8970
0.8864
0.8964
Neural Network
Logistic Regression (Final)
Random Forest (Final)
Model


Confusion Matrix – Logistic Regression (Final Model)
True Label
Low Obesity
High Obesity
47.0
53.0
8.4
91.6
Low Obesity
High Obesity
Predicted Label
Percentage
90
80
70
60
50
40
30
20
10

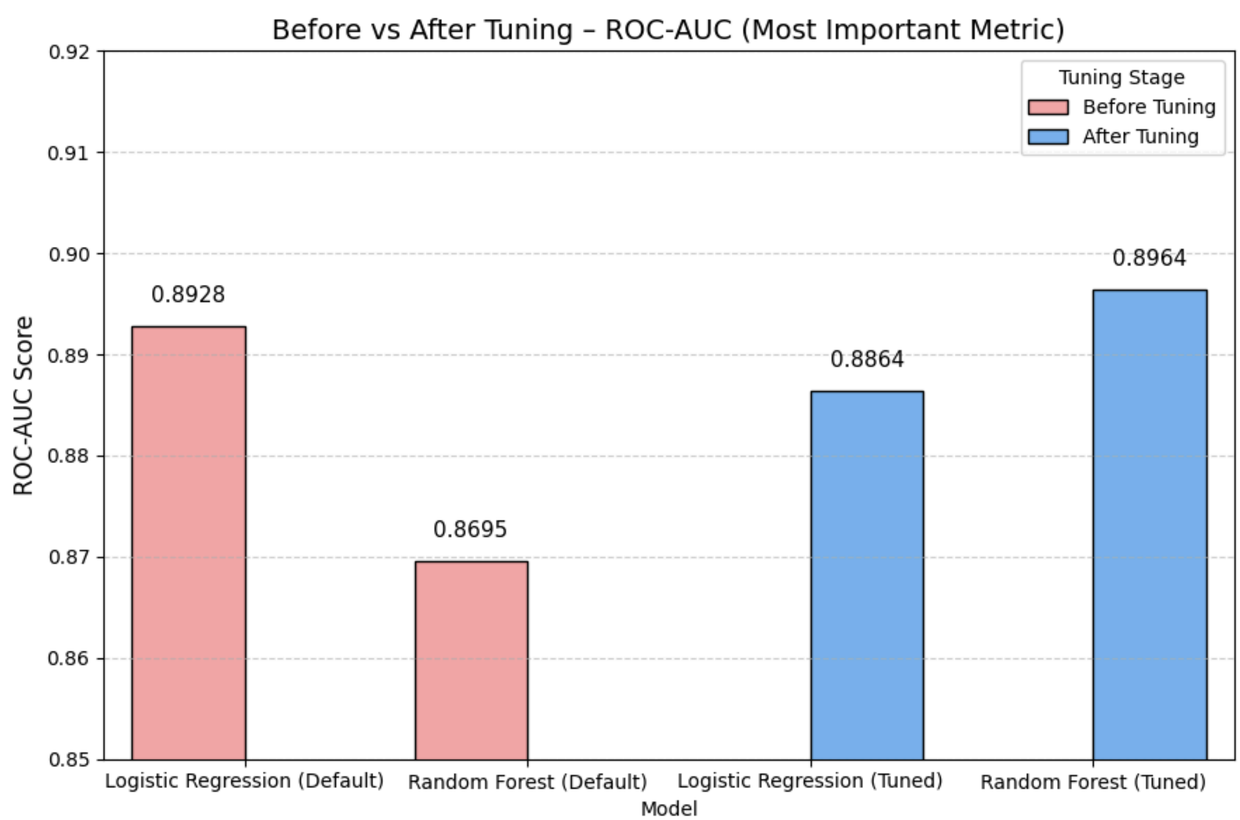
4.2 Impact of Hyperparameter Tuning and Feature Selection

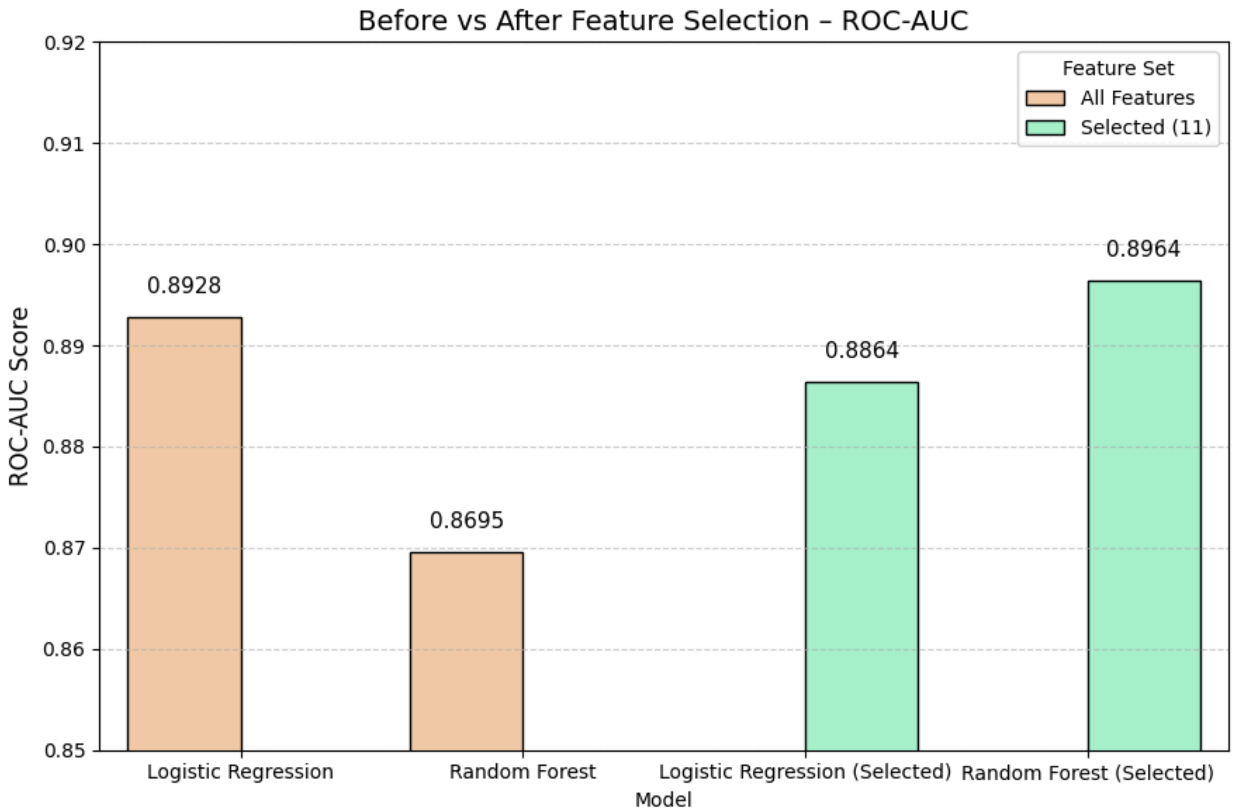
Tuning improved Logistic Regression (better ROC-AUC).

Feature selection to 11 features made models simpler and faster.

Performance dropped a little but stayed useful.

Tuning and selection helped avoid overfitting and improve generalization.





4.3 Interpretation of Results

Sample_Size, income, education, race, age, and some states are main factors for obesity.

Higher income and education groups have lower obesity.

This matches real-world knowledge.

4.4 Limitations

- Dataset is only from USA
- Many rows had missing stratification data
- Models may not work perfectly on new countries

4.5 Suggestions for Future Research

- Use more advanced models
- Add time-series analysis
- Do feature engineering
- Test on larger or international data

5. References

1. Centers for Disease Control and Prevention (CDC). Nutrition, Physical Activity, and Obesity – BRFSS. https://data.cdc.gov
2. Scikit-learn documentation
3. PyTorch documentation
4. United Nations Sustainable Development Goals